Science Syllabus

Grades 1–4

2026–2027

INSTRUCTOR INFORMATION

Instructor	Email	Class Days
Tiffany Young	Tiffany@rivertech.me	Tuesday
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COURSE OVERVIEW

The Grades 1–4 Science program develops students' understanding of the natural and physical world through reading, observation, discussion, hands-on experiments, scientific vocabulary, and practical application.

Students use Evan-Moor Skill Sharpeners Science Workbooks at their grade level. The curriculum introduces age-appropriate concepts in physical science, life science, earth and space science, environmental science, and engineering. Units combine informational reading, key vocabulary, comprehension activities, visual learning, and hands-on activities to strengthen understanding of scientific concepts.

Generation Genius educational videos may be used when they fit the lesson as an engaging visual reinforcement of concepts being taught.

Practical experiments and hands-on activities described in the Evan-Moor curriculum are incorporated throughout the course whenever appropriate. Students learn by observing, testing, comparing, discussing, recording results, and connecting scientific ideas to real-world experiences.

The overall goal is to help students become curious observers, thoughtful questioners, and confident young scientists who can understand basic scientific concepts and explain what they observe.

COURSE MATERIALS

Required Materials

- Evan-Moor Skill Sharpeners Science Workbooks (at the student's grade level)
- Generation Genius educational videos, used as reinforcement when they fit the lesson

SYLLABUS

Scientific Inquiry & Skills

Throughout Grades 1–4, students practice foundational scientific habits and skills, including observing and describing, asking questions, making predictions, comparing and classifying, using scientific vocabulary, following steps in an investigation, recording observations, discussing results, drawing simple conclusions, and connecting scientific ideas to everyday life. As students grow older, they are increasingly encouraged to explain why they think something happened and to support their ideas with observations and evidence.

Grade 1 Science

Grade 1 introduces foundational concepts in physical, life, earth and space, environmental, and engineering science. Topics include sound, light, shadows, technology, leaves, animals and eggs, mothers and babies, birds, insects, the night sky, the sun, seasons, uses of wood, earth materials, recycling, tools, and how tools are designed for different purposes. Students learn through short readings, pictures, discussion, observation, and hands-on projects.

Grade 2 Science

Grade 2 expands students' understanding of matter, forces, living things, earth systems, space, and engineering. Topics include matter and changes in matter; forces and motion; the human body; animal life cycles; plants and habitats; space and the sun; weather; geology; water and ice; and basic engineering concepts. Hands-on activities help students observe how materials change, how forces work, how living things grow, and how earth and weather systems behave.

Grade 3 Science

Grade 3 develops understanding of forces, life cycles, ecosystems, fossils, weather, and natural events. Topics include forces and motion, magnets, gravity, life cycles, growth and change, animal groups and offspring, fossils and dinosaurs, adaptations, ecosystems, weather and climate, hurricanes, tornadoes, and floods. Students make increasingly strong connections between cause and effect, environmental conditions, adaptation, and change.

Grade 4 Science

Grade 4 develops students' understanding of the human body, earth processes, waves, and energy. Topics include muscles and bones, skin, eyesight, hearing, weathering and erosion, glaciers, earthquakes, volcanoes, waves in water, sound waves, digital waves, heat energy, electrical energy, and light energy. Students are increasingly encouraged to explain scientific relationships, interpret observations, and connect scientific concepts to real-life applications.

Hands-On Experiments & Practical Science

Hands-on learning is an important part of the course. Students participate in experiments, demonstrations, models, and practical activities described in the Evan-Moor curriculum. Activities may

include investigating light and shadows, testing forces, observing plant growth, modeling erosion, exploring sound and waves, examining materials, or studying energy. Students practice making predictions, observing changes, comparing results, using materials safely, discussing what happened, and connecting scientific ideas to real-world experience.

Generation Genius Reinforcement

Generation Genius educational videos may be used when they directly support the topic being studied. Videos are used as reinforcement rather than as the primary curriculum. They may introduce or review a concept, provide a visual explanation, or help students observe scientific processes that are difficult to reproduce in the classroom. Teachers may pause videos for questions, discussion, prediction, and clarification.

Science Vocabulary & Reading

Science instruction also strengthens reading and language skills. Students learn grade-appropriate scientific vocabulary, read short informational passages, interpret illustrations and diagrams, answer comprehension questions, and discuss science concepts orally. As students progress, they are increasingly expected to use scientific vocabulary correctly and explain concepts in their own words.

Discussion & Communication

Verbal discussion is incorporated throughout science instruction. Students are encouraged to ask questions, share observations, explain what they notice, make predictions, describe cause and effect, compare ideas, explain experiment results, and use scientific vocabulary in conversation. This develops scientific thinking and communication rather than simple memorization of facts.

Individualized Learning

Science instruction is adjusted according to students' age, reading level, background knowledge, and readiness. Teachers may simplify explanations, provide extra demonstrations, repeat activities, or offer additional reading and discussion as needed. Students ready for greater challenge may explore topics more deeply or complete additional investigations. The focus is on understanding and curiosity, not simply completing workbook pages.

COURSE GOALS

By the end of the elementary Science program, students will demonstrate continued growth in scientific observation, vocabulary, questioning, prediction, experimentation, comprehension, cause-and-effect thinking, scientific discussion, and understanding of physical, life, earth, environmental, and engineering concepts.

Students will increasingly be able to observe the world carefully, ask meaningful questions, explain basic scientific ideas, participate in simple investigations, and connect what they learn to everyday life.